Hirona Jacqueline Arai

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Ph.D. candidate in NLP (USC INK Lab, advisor Xiang Ren) working on pragmatic interpretation in LLMs: how models recover meaning that depends on speaker intent, social cues, and context rather than surface form. First-author paper at ACL 2026; co-author at ICLR and EMNLP. Background in computer science and linguistics. Seeking a Summer 2027 research internship.

EDUCATION

University of Southern California, Los Angeles, CA — Sep 2022 – May 2028 (expected)
Ph.D. Candidate, Computer Science, Viterbi School of Engineering — GPA 3.85/4.0
Advisor: Xiang Ren, Intelligence and Knowledge Discovery (INK) Lab. Coursework: Advanced NLP, Science of LLMs, Natural Language Dialogue Systems, Semantics, Advanced Analysis of Algorithms.

Middlebury College, Middlebury, VT — Sep 2018 – May 2022
B.A. Computer Science (Highest Honors), Minor in Linguistics — summa cum laude, GPA 3.9/4.0

RESEARCH EXPERIENCE

Graduate Research Assistant — INK Lab, University of Southern California — Sep 2022 – Present

- **DRInQ (ACL 2026, first author):** built a benchmark for pragmatic reasoning about conversational implicature in questions that isolates pragmatic variation while holding surface form fixed, with a semi-automated pipeline generating question–context–interpretation instances at scale.
- Found a consistent generation–inference asymmetry in state-of-the-art LLMs (they generate plausible pragmatic scenarios when guided but fail to recover the intended implication at inference), showed structured prompting improves smaller models' alignment with human judgments, and ran a human-vs-model comparative writing study.
- **Underspecified editing instructions (in progress):** developing a property-level framework for how LLMs operationalize requests like "make this simpler"; built an LLM-annotation pipeline over 200K+ WildChat conversations (~30K validated edit requests), an evidence-grounded LLM-as-judge for property change, and a controlled dataset for human evaluation.
- Co-author on procedural-knowledge distillation for small LMs with AI2 (PlaSma, ICLR 2024), factual recall from visual references in VLMs (Findings of EMNLP 2025), and translation of culturally untranslatable terms (Findings of EMNLP 2026).

Research Assistant — Department of Computer Science, Middlebury College (advisor: John Foley) Jun 2021 – Dec 2021

- Trained and analyzed neural models to detect literary devices in large poetry archives to improve search.
- Senior thesis: built a silver-standard corpus from Wikipedia text via data augmentation to train a part-of-speech tagger robust to less-formal text (poetry, internet speech).

Research Intern — AIST (Natl. Institute of Advanced Industrial Science and Technology), Tokyo Jun 2020 – Aug 2020
Artificial Intelligence Research Center (AIRC-KIRT); advisor: Hiroya Takamura

- Measured semantic change in English produced by language learners using feature learning and embedding models.
- Designed and contributed to an English/Japanese cross-linguistic study of lexical variation.

PUBLICATIONS

- **Hirona J. Arai** and Xiang Ren. *DRInQ: Evaluating Conversational Implicature with Controlled Context Variation.* ACL 2026.
- Julian Maynes, Nairi Zeytounzian, **Hirona J. Arai**, Xiang Ren. *NoHurt: Context-Sensitive Benchmarking for Quantifying Information Acquiescence During High-Risk Dialogue.* Under review, ACL Rolling Review, Oct 2026.
- Jacob Bremerman, Brihi Joshi, **Hirona J. Arai**, Jonathan May, Xiang Ren. *Translating the Untranslatable.* Findings of EMNLP 2026.
- Dhananjay Ashok, Ashutosh Chaubey, **Hirona J. Arai**, Jonathan May, Jesse Thomason. *Can VLMs Recall Factual Associations From Visual References?* Findings of EMNLP 2025.

- Faeze Brahman, Chandra Bhagavatula, Valentina Pyatkin, Jena D. Hwang, Xiang Lorraine Li, **Hirona J. Arai**, Soumya Sanyal, Keisuke Sakaguchi, Xiang Ren, Yejin Choi. *PlaSma: Making Small Language Models Better Procedural Knowledge Models for (Counterfactual) Planning.* ICLR 2024.

TEACHING AND MENTORING

Head Teaching Assistant — CSCI 544 Applied Natural Language Processing, USC (Prof. Xiang Ren) Aug 2026 – Dec 2026
- Lead a team of 4 TAs for a 300-student master's course; deliver the PyTorch tutorial lecture (second year); run course logistics and office hours, and mentor and grade team projects from proposal to final report.

Teaching Assistant — CSCI 444 Natural Language Processing, USC (Prof. Xiang Ren) Jan 2026 – May 2026
- Sole TA for a 30-student undergraduate course: logistics, office hours, project mentoring, and grading.

Research Mentor — INK Lab, USC Jun 2023 – Present
- Mentored six students from high school through master's level across five research projects, adapting scope and supervision to each student's experience; two projects have led to papers (one under review, one in preparation).
- **2026:** guided two CSCI 444 undergraduates in turning their course project into a paper (NoHurt, under review at ACL Rolling Review, Oct 2026); co-mentoring a master's student with Sahana Ramnath on hallucinations in explanations for authorship verification, targeting a student research workshop (EACL/NAACL).
- **Spring 2024 – Fall 2025:** supervised an undergraduate research assistant on the early stages of the DRinQ project.
- **Summer 2023:** mentored two Los Angeles high-school students through independent research projects in USC's SHINE+ program.

Computer Science Tutor — Middlebury College 2020 – 2022
- Tutored Programming Languages, Software Development, and Data Structures; President of the Linguistics Club.

SKILLS

Technical: Python, PyTorch, Hugging Face Transformers, LLM APIs, Slurm/HPC; benchmark design, LLM-as-judge and annotation pipelines, human evaluation
Languages: English (native), Japanese (native)

HONORS AND AWARDS

- Yanai Tadashi Foundation Scholarship, 2018 – 2022 (merit-based, debt-free award for Japanese students at top U.S. universities, founded by UNIQLO's Tadashi Yanai).
- Highest Honors in Computer Science and summa cum laude, Middlebury College, 2022.
- Breakthrough Junior Challenge: educational video on rechargeable batteries, 43K+ views.